

Name: Monisha Raghini Chelle

RollNo:2520030537

Section-6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

14.To Parse Append Operator, Open Files in Append Mode, Redirect Streams, Preserve Existing Data, Validate Appends, Test Scenarios.

To Parse stderr Redirection, Redirect Error Streams, Capture Error Output, Verify File Contents, Handle Failures, Test Error Cases.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c hello:Zone.Identifier    hello_shortcut skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create the Append Redirection Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano append_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 append_redirection.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define INPUT_SIZE 200

int main()
{
    char input[INPUT_SIZE];
    int saved_stdout;
    int fd;

    printf("Append Redirection Program\n");
    printf("Enter a command in the form: echo Hello >> output.txt\n");
    printf("Type exit to quit.\n");

    while (1)
    {
        printf("append> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        char *redirect = strstr(input, ">>");

        if (redirect == NULL)
        {
            printf("Error: Append operator '>>' not found.\n");
            continue;
        }

        *redirect = '\0';

        char *command = input;
        char *filename = redirect + 2;

        while (*command == ' ')
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 append_redirection.c *
while (*command == ' ')
    command++;

while (*filename == ' ')
    filename++;

if (*command == '\\0')
{
    printf("Error: Command is missing.\\n");
    continue;
}

if (*filename == '\\0')
{
    printf("Error: Output file name is missing.\\n");
    continue;
}

fd = open(filename, O_WRONLY | O_CREAT | O_APPEND, 0644);

if (fd == -1)
{
    perror("open");
    printf("Error: Cannot open output file: %s\\n", filename);
    continue;
}

saved_stdout = dup(STDOUT_FILENO);

if (saved_stdout == -1)
{
    perror("dup");
    close(fd);
    continue;
}

if (dup2(fd, STDOUT_FILENO) == -1)
{
    perror("dup2");
    close(fd);
    close(saved_stdout);
    continue;
}

close(fd);

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 append_redirection.c *

    if (saved_stdout == -1)
    {
        perror("dup");
        close(fd);
        continue;
    }

    if (dup2(fd, STDOUT_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        close(saved_stdout);
        continue;
    }

    close(fd);

    if (strncmp(command, "echo ", 5) == 0)
    {
        printf("%s\n", command + 5);
    }
    else
    {
        printf("%s\n", command);
    }

    fflush(stdout);

    if (dup2(saved_stdout, STDOUT_FILENO) == -1)
    {
        perror("dup2 restore");
    }

    close(saved_stdout);

    printf("Data appended to: %s\n", filename);
    printf("Output stream restored successfully.\n");
}

printf("Program exited successfully.\n");

return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo

Save the file:

Press Ctrl + O

Press Enter

Press Ctrl + X

```
Write to File: append_redirection.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse  
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

Compile:

gcc append_redirection.c -o append_redirection

Verify:

ls -l append_redirection.c append_redirection

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano append_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc append_redirection.c -o append_redirection
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l append_redirection.c append_redirection
-rwxr-xr-x 1 monis monis 16608 Sep 18 12:30 append_redirection
-rw-r--r-- 1 monis monis 2374 Sep 18 12:29 append_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 3 — Create Initial Data for the Append Test

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano append.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

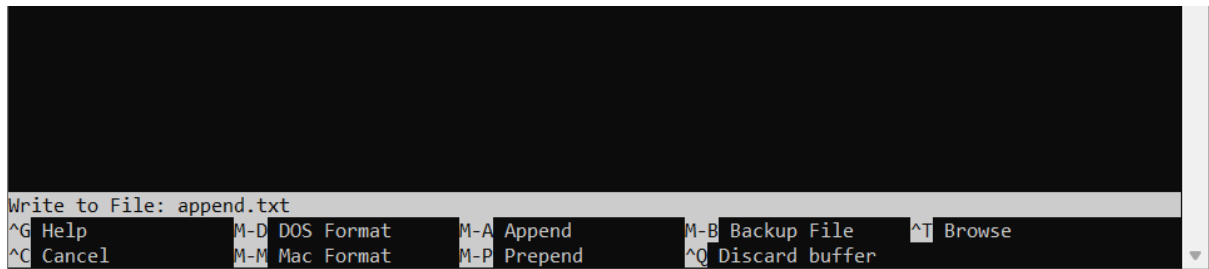
```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 append.txt *
Original data
```

Save:

Ctrl + O

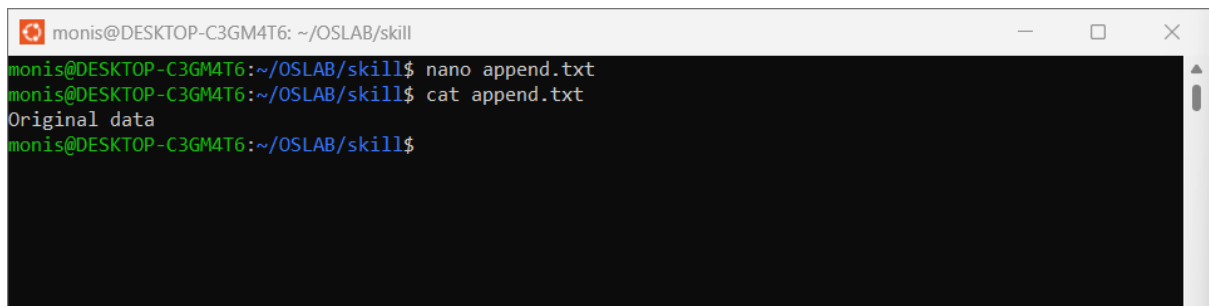
Enter

Ctrl + X



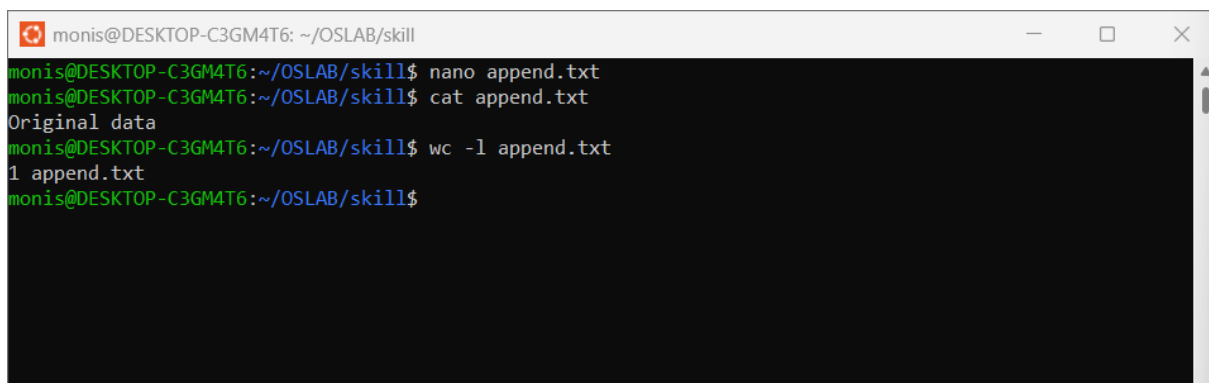
Now verify:

cat append.txt



Now check the file before appending:

wc -l append.txt



Step 4 — Test Append Redirection (>>)

Now run the program:

./append_redirection

At the prompt, enter:

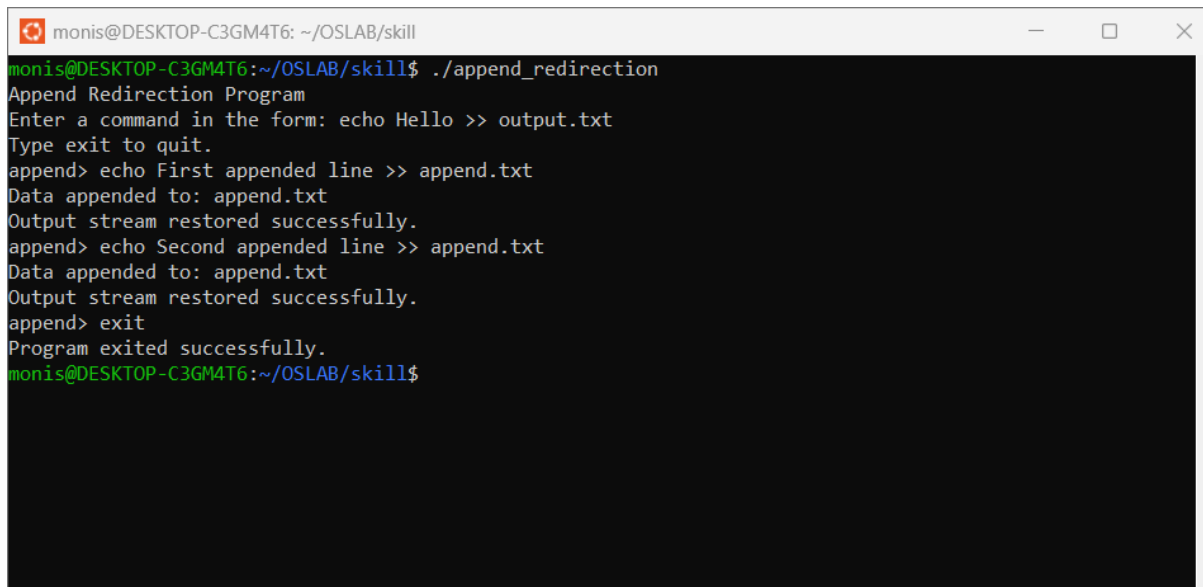
echo First appended line >> append.txt

Now append a second line:

echo Second appended line >> append.txt

Now exit:

exit



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./append_redirection
Append Redirection Program
Enter a command in the form: echo Hello >> output.txt
Type exit to quit.
append> echo First appended line >> append.txt
Data appended to: append.txt
Output stream restored successfully.
append> echo Second appended line >> append.txt
Data appended to: append.txt
Output stream restored successfully.
append> exit
Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

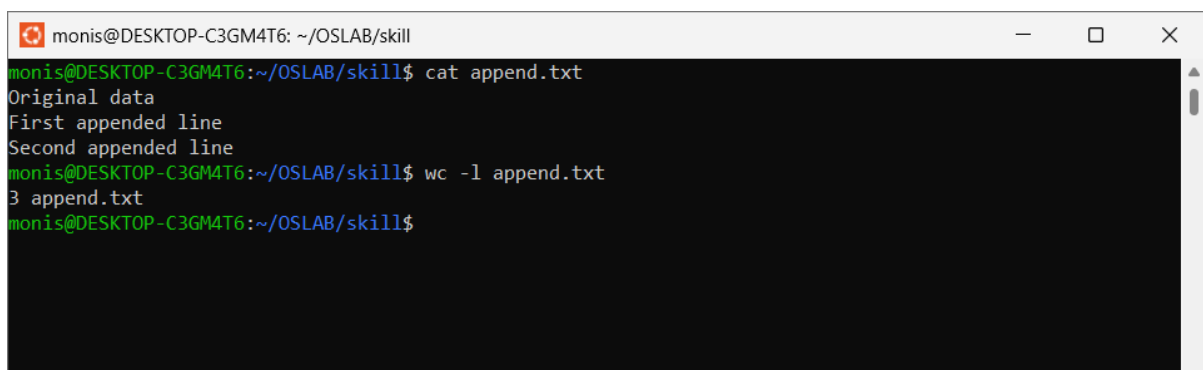
Verify that existing data was preserved

Run:

cat append.txt

Then verify the number of lines:

wc -l append.txt



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat append.txt
Original data
First appended line
Second appended line
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ wc -l append.txt
3 append.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Create the stderr Redirection Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano stderr_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
GNU nano 8.7.1 stderr_redirection.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define INPUT_SIZE 200

int main()
{
    char input[INPUT_SIZE];
    int saved_stderr;
    int fd;

    printf("stderr Redirection Program\n");
    printf("Enter a command in the form: ls missing_file 2> error.txt\n");
    printf("Type exit to quit.\n");

    while (1)
    {
        printf("stderr> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        char *redirect = strstr(input, "2>");

        if (redirect == NULL)
        {
            printf("Error: stderr redirection operator '2>' not found.\n");
            continue;
        }

        *redirect = '\0';

        char *command = input;
        char *filename = redirect + 2;

        while (*command == ' ')
    }
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^V Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 stderr_redirection.c *

    command++;

    while (*filename == ' ')
        filename++;

    if (*command == '\0')
    {
        printf("Error: Command is missing.\n");
        continue;
    }

    if (*filename == '\0')
    {
        printf("Error: Error output file name is missing.\n");
        continue;
    }

    fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (fd == -1)
    {
        perror("open");
        printf("Error: Cannot open error output file: %s\n", filename);
        continue;
    }

    saved_stderr = dup(STDERR_FILENO);

    if (saved_stderr == -1)
    {
        perror("dup");
        close(fd);
        continue;
    }

    if (dup2(fd, STDERR_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        close(saved_stderr);
        continue;
    }

    close(fd);

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 stderr_redirection.c *
    close(saved_stderr);
    continue;
}

close(fd);

if (strncmp(command, "ls ", 3) == 0)
{
    char *filename_to_open = command + 3;

    FILE *file = fopen(filename_to_open, "r");

    if (file == NULL)
    {
        perror("ls");
    }
    else
    {
        printf("File exists: %s\n", filename_to_open);
        fclose(file);
    }
}
else
{
    fprintf(stderr, "Command failed: %s\n", command);
}

fflush(stderr);

if (dup2(saved_stderr, STDERR_FILENO) == -1)
{
    perror("dup2 restore");
}

close(saved_stderr);

printf("Error output redirected to: %s\n", filename);
printf("stderr stream restored successfully.\n");
}

printf("Program exited successfully.\n");

return 0;
}

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

Save:

Ctrl + O

Enter

Ctrl + X

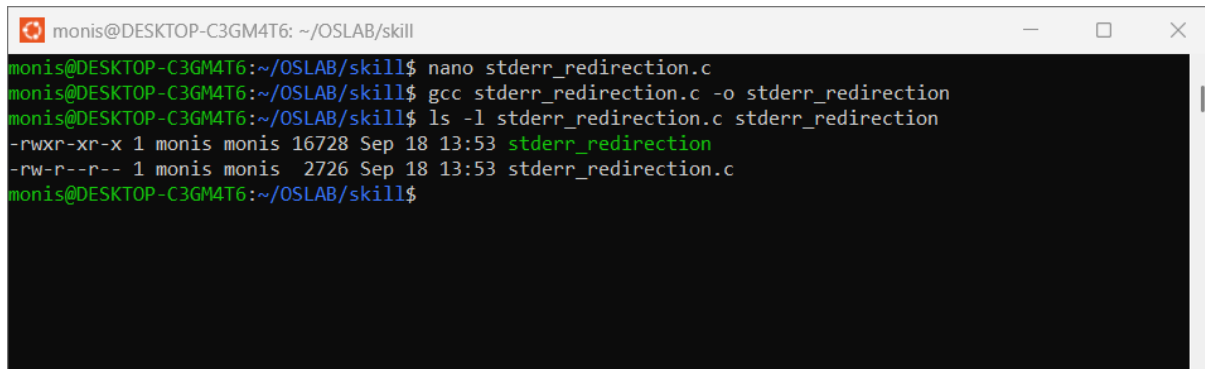
```
}
Write to File: stderr_redirection.c
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel     M-M Mac Format  M-P Prepend    ^O Discard buffer
```

Compile:

gcc stderr_redirection.c -o stderr_redirection

Verify compilation:

ls -l stderr_redirection.c stderr_redirection

A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill' showing the following commands and output:

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano stderr_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc stderr_redirection.c -o stderr_redirection
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l stderr_redirection.c stderr_redirection
-rwxr-xr-x 1 monis monis 16728 Sep 18 13:53 stderr_redirection
-rw-r--r-- 1 monis monis 2726 Sep 18 13:53 stderr_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 6 — Test stderr Redirection

Run:

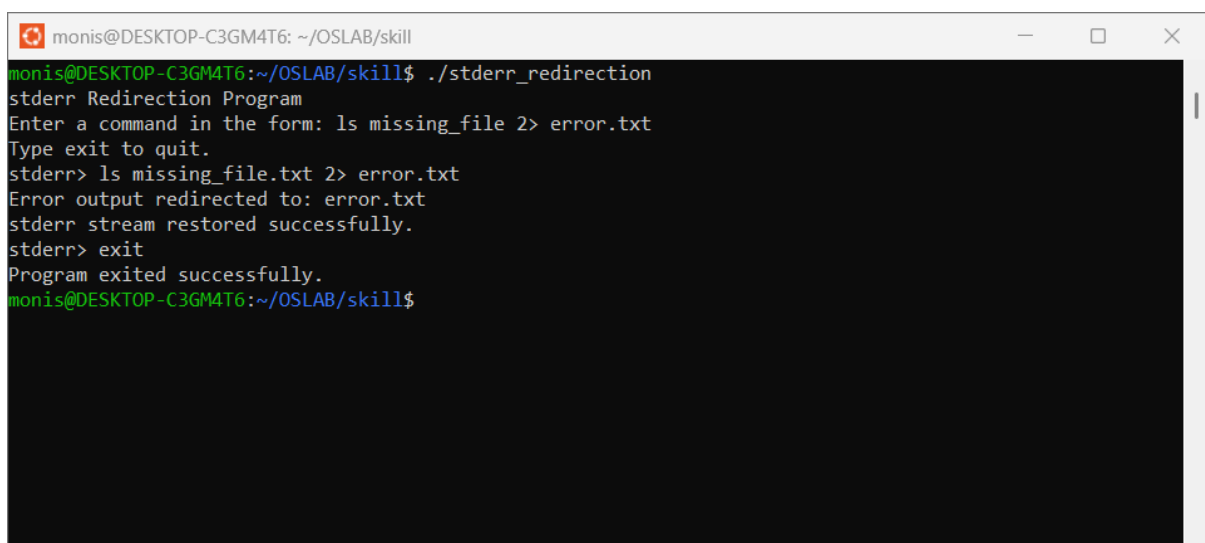
./stderr_redirection

Now intentionally use a file that does not exist:

ls missing_file.txt 2> error.txt

Now exit:

exit

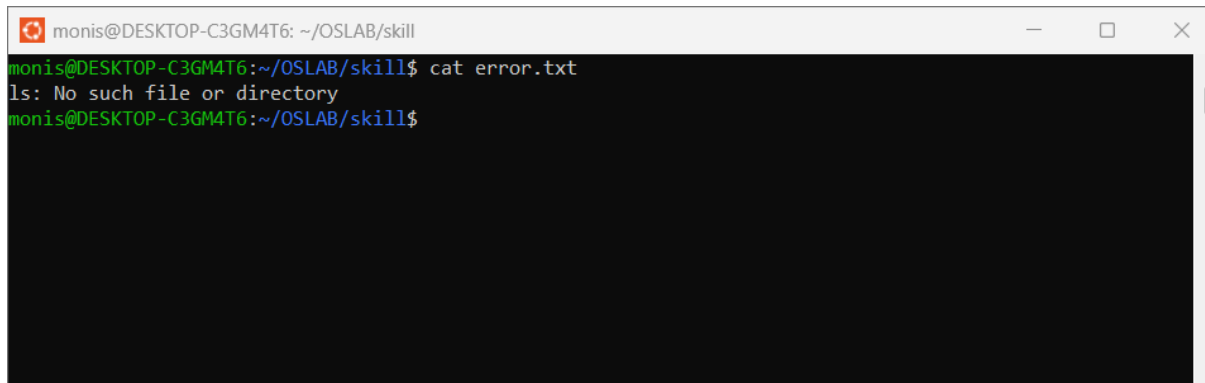
A terminal window titled 'monis@DESKTOP-C3GM4T6: ~/OSLAB/skill' showing the following commands and output:

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./stderr_redirection
stderr Redirection Program
Enter a command in the form: ls missing_file 2> error.txt
Type exit to quit.
stderr> ls missing_file.txt 2> error.txt
Error output redirected to: error.txt
stderr stream restored successfully.
stderr> exit
Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Verify the captured error

Run:

`cat error.txt`



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat error.txt
ls: No such file or directory
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

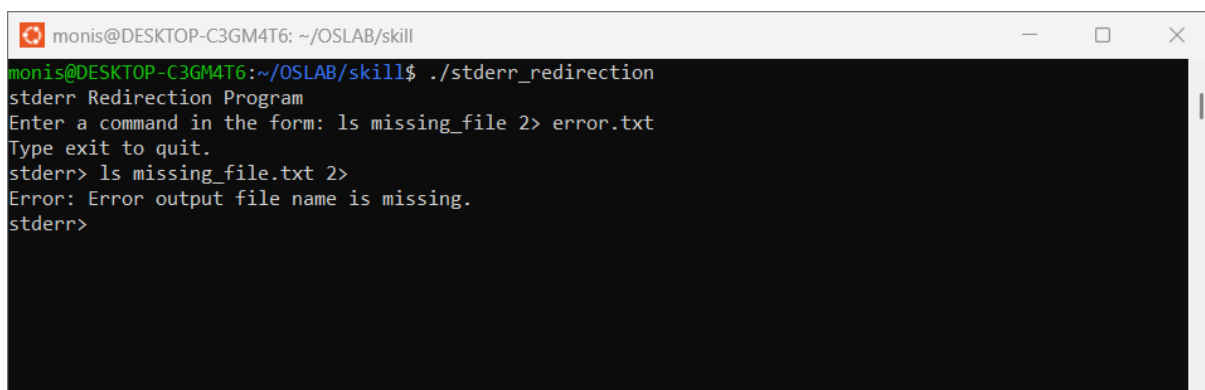
Step 7 — Test the stderr Error Case

Run:

`./stderr_redirection`

At the prompt, enter:

`ls missing_file.txt 2>`



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./stderr_redirection
stderr Redirection Program
Enter a command in the form: ls missing_file 2> error.txt
Type exit to quit.
stderr> ls missing_file.txt 2>
Error: Error output file name is missing.
stderr>
```

Now test an input without the 2> operator:

`ls missing_file.txt`

Finally:

`exit`

